Proximal Policy Optimisation for Multi-Asset ETF Allocation under Transaction Costs

SIQuant Research[1]

[1]Hypothesis × Quant

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Abstract

Hypothesis: H_0: the Sharpe ratio of a PPO allocation policy does not exceed that of daily-rebalanced equal weight; H_1: it does

Data: Total returns of SPY, TLT, GLD, EEM, HYG; training before 2023, out-of-sample 2023-02-01 to 2026-10-22 (935 days)

Method: PPO (3 seeds, 120,000 steps each, averaged), softmax weights, 5 bps cost per unit turnover, decision at close t applied to day t+1, one-sided stationary-bootstrap test of the Sharpe difference

Results: PPO Sharpe 0.18 vs equal weight 1.17 (difference -0.99, $p = 1.000$).

Research decision: REJECTED

1 Design

The environment exposes the last 20 daily returns of each asset (scaled by training volatility) and the current drifted weights. The agent outputs unconstrained scores mapped to long-only fully invested weights by a softmax; reward is the log net return. Training episodes are random one-year windows before 2023; the test period is simulated once and never used for tuning [Moody and Saffell, 2001].

2 Results

Table 1: Out-of-sample performance net of 5 bps cost per unit turnover

Strategy	Ann. return	Ann. vol.	Sharpe	Max drawdown
PPO ensemble	1.34%	11.16%	0.18	-15.1%
Equal weight	12.05%	10.19%	1.17	-8.7%
SPY buy-and-hold	20.06%	14.82%	1.31	-18.8%
60/40 SPY/TLT	10.60%	11.08%	0.96	-12.7%

The ensemble policy turns over 64.4% of the portfolio per day on average.

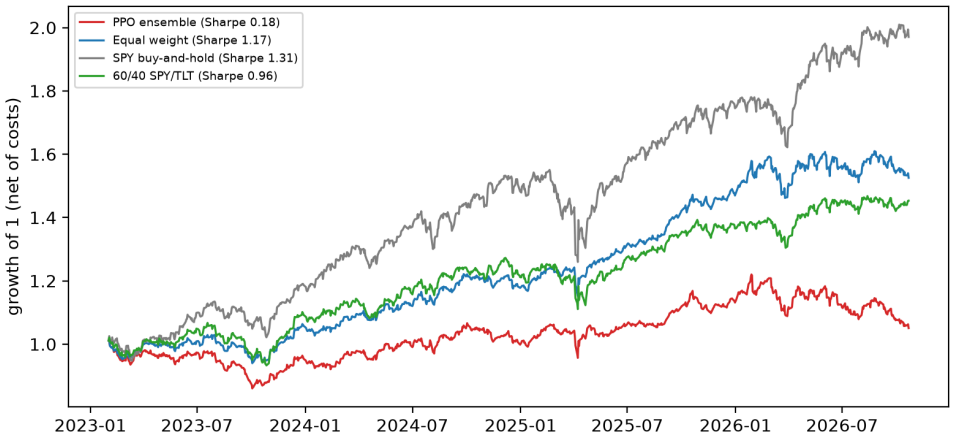


Figure 1: Out-of-sample growth of one unit.

3 Limitations

Five ETFs and a single three-year test window give low statistical power; cash and financing are ignored; PPO hyper-parameters were fixed a priori, not searched, and results vary across seeds (averaged here).

References

John Moody and Matthew Saffell. Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks*, 12(4):875–889, 2001.